



Unsupervised learning

Exploring the **unknown** with unsupervised learning.

5

WEEKS

175

HOURS

7

LESSONS

Unsupervised learning refers to machine learning techniques used to analyse and cluster unlabelled data. In this module, we'll examine how to approach **data exploration without explicit guidance** from the dataset.

We'll introduce techniques such as **principal component analysis (PCA)** and other **dimensionality reduction techniques**, which allow us to make sense of large datasets without the time costs normally associated with analysing them. We also explore the world of **clustering**, with K-means and hierarchical techniques, finding insights in seemingly chaotic data structures. We also look at **Gaussian mixture models** before introducing **geospatial analysis**. Finally, we'll investigate the workings of **recommender systems**, a machine learning technique that helps to narrow down and predict user preferences and choices.

This module consists of **multiple videos**, **examples**, and **practical assessments**, all of which explain and reinforce our understanding of these machine learning techniques.

Learning activities

Participating in various learning activities will **enhance our understanding** of unsupervised learning techniques and help us **develop a skillset** that can be applied to a **variety of data problems**.

We learn by doing. We'll work on practical problem-solving and real-world projects.

Learn

Watch animated videos and explore practical examples to learn NLP and classification concepts.



7

Sketch videos



1

Slide deck



8

Examples



7

Additional
resource activities

Apply

Practise unsupervised learning techniques by following detailed step-by-step guides and applying these techniques to real-world scenarios.



7

Exercises

Assess

Test and track your understanding of clustering and other unsupervised learning techniques.



8

KQ assessments



3

MCQ assessments



1

Practical project



1

Exam



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Week 1

Lesson: Principal component analysis

In this lesson, we begin by exploring the concept of **unsupervised learning** and its relevance in extracting meaningful **patterns from unlabeled data**. We are then introduced to the concept of dimensionality reduction, where we examine the challenges of working with high-dimensional data and the benefits of reducing the number of features.

We will look at a specific dimensionality reduction technique, **principal component analysis (PCA)**, and discuss how it transforms high-dimensional data into a lower-dimensional space while preserving as much information as possible.

- ✓ Understand the idea of **unsupervised learning** and how it differs from supervised learning.
- ✓ Gain insight into the concept of **dimensionality reduction** and its importance in **simplifying high-dimensional data**.
- ✓ Learn the theory behind **principal component analysis (PCA)** as a dimensionality reduction technique.
- ✓ Implement PCA on a dataset using the **Python sklearn library**.

Lesson: Advanced dimensionality reduction

Dimensionality reduction techniques like multi-dimensional scaling (MDS) and t-distributed Stochastic Neighbour Embedding (t-SNE) are essential tools in data science, enabling practitioners to **navigate high-dimensional spaces** effectively for analysis and visualisation.

In this lesson, we'll explore **advanced dimensionality reduction techniques**, including **MDS** and **t-SNE**. We'll review Principal Component Analysis (**PCA**), introduce **MDS** for preserving distances in reduced dimensions, explain t-SNE for visually distinguishing complex datasets, and implement these techniques on image data to uncover underlying patterns.

- ✓ Understand how to use multi-dimensional scaling (**MDS**) to **preserve distances in reduced dimensions**.
- ✓ Explain t-distributed Stochastic Neighbour Embedding (**t-SNE**) for **visually distinguishing complex datasets** in lower-dimensional spaces.
- ✓ Demonstrate implementing **dimensionality reduction techniques** on image data to **visualise patterns**.

Week 2

Lesson: K-means clustering

In this lesson, we will cover the fundamentals of clustering and explore the implementation of the **K-means clustering algorithm**, a key technique in unsupervised learning. We will gain an understanding of how clustering algorithms **group data points based on similarities**, allowing for the identification of patterns and structures within datasets. Through a combination of theoretical explanations and practical demonstrations, we will learn how to apply K-means clustering using sklearn and explore methods for determining the optimal number of clusters.

- ✓ Explain the **concept of clustering** and its significance in data analysis.
- ✓ Implement the **K-means clustering algorithm in Python** using **sklearn**.
- ✓ Evaluate **different techniques for determining the optimal number** of clusters in K-means clustering.

Lesson: Hierarchical clustering

Hierarchical clustering is a method of cluster analysis used in data science. It facilitates the exploration of complex datasets by organising them into a **hierarchical structure**. Through this method, **data points are grouped together based on their similarities**, forming clusters that can be further subdivided into smaller clusters.

In this module, we will explore hierarchical clustering, covering its **algorithm** and the **dendrogram interpretation**. We will **implement** this method using sklearn and SciPy, and learn how to determine the optimal **number of clusters**.

- ✓ Understand **hierarchical clustering**.
- ✓ Interpret **dendrograms**.
- ✓ Implement **agglomerative clustering** in sklearn and SciPy.
- ✓ Determine the **optimal number** of clusters from a **dendrogram**.

Week 3

Lesson: Gaussian mixture models

Gaussian mixture models (GMMs) are sophisticated unsupervised machine learning tools, used to uncover **latent structures** within data. Unlike more simplistic clustering techniques, GMMs accommodate **mixed membership** of data points to **multiple clusters**, making them highly effective for **complex data distributions** involving **overlapping clusters** and **non-spherical shapes**.

In this lesson, we'll delve into the **theory and application of Gaussian mixture models**. We'll start by **comparing GMMs with simpler clustering methods** like k-means to highlight their advantages in soft clustering scenarios. Following that, we will explore the **mathematical foundation of GMMs**, including the Expectation-Maximisation (EM) algorithm, which plays a crucial role in estimating the parameters of GMMs.

- Describe the **concept of mixture models** and how GMMs represent a **probabilistic model** for clustering.
- Differentiate GMMs from k-means clustering in terms of **cluster assignment** and **flexibility**.
- Explain the **Expectation-Maximisation algorithm** and its importance in **optimising GMM parameters**.
- Implement GMMs using **Scikit-learn** to perform clustering on **real-world datasets**.

Lesson: Clustering and geospatial analysis

Clustering and geospatial analysis are critical components of data science, particularly in understanding **spatial relationships and patterns** within data. By applying clustering techniques to geospatial data, analysts can **uncover hidden structures** and make **meaningful interpretations of geographic phenomena**. This involves processing large datasets, employing statistical methods to group similar data points, and visualising complex geospatial information in a comprehensible manner.

In this lesson, we'll look at **practical aspects of clustering and geospatial analysis**. We'll start by learning how to **process and cluster census data**, focusing on identifying patterns and relationships. Using the Python library **GeoPandas**, we'll introduce the essentials of **geospatial data plotting**, including how to manipulate attributes like size and colour to create multidimensional visual representations.

- Process** and **cluster census data** to uncover patterns.
- Utilise **GeoPandas** for **plotting** and **manipulating** geospatial data.
- Create **multidimensional plots** adjusting attributes like size and colour to reveal deeper insights.
- Understand and apply **various clustering techniques** to geospatial datasets.
- Interpret and utilise **external resources to extend knowledge** and application of geospatial analysis techniques.

Week 4

Lesson: Recommender systems

Recommender systems are vital in data science, providing **personalised recommendations**. They use **collaborative** and **content-based filtering** to analyse user behaviour or item attributes, recommending relevant items.

In this lesson, we'll explore recommender systems, focusing on collaborative and content-based filtering. We'll understand **user similarity**, **utility matrices**, and the "**cold-start problem**." Implementing a collaborative filtering algorithm, we'll learn to compute user similarity matrices and generate recommendations. Contrasting with content-based filtering, we'll discern trade-offs for recommendation variety.

- Gain an **intuition** for the basic operation of **content** and **collaborative filtering**.
- Implement a simple **content** and **collaborative-based filtering**.
- Understand some of the **trade-offs** offered between **content** and **collaborative-based filtering**.

Practical project: **Unsupervised learning techniques**
It's time to put our **newly acquired skills to the test** in a practical project!

Week 5

Exam: **Unsupervised learning**

It is time to review all the work that's been covered up to now and **test our knowledge**. Make sure to **cover each week's section in detail** before attempting this exam! It will be a combination of **theoretical** and **practical questions**, aimed at testing the **general understanding of concepts**, as well as the **application** and **interpretation of our new skills**.

Module summary

The purpose of unsupervised learning is to assist in **finding patterns** or **clusters** in large datasets. In this module, we covered a variety of techniques that can be used to help us do this. We looked at how to **reduce the size of the feature space** through **principal component analysis**, as well as **other dimensionality reduction techniques**. We examined both **k-means** and **hierarchical clustering**, before digging into **Gaussian mix models** and **recommender systems**. The reality of working with data is that it's not always clean, small or well-structured.

The techniques introduced in this unsupervised learning module should **help us deal with this lack of structure** and **make the process of analysing and gathering insights** from the data **more manageable**.

What's next?

Throughout this course, we've introduced a large number of **machine learning algorithms** and methods that can be **used when working with data of various shapes and forms**. One of the most important factors to remember is to **choose the best method to solve a problem**, rather than one that appears to be very smart and likely to impress peers.

Sometimes simple is best, being able to **explain what our code is doing** and **how it is getting to the answer** can be as important as solving the problem itself! It is now time for you to apply your knowledge to problems that matter to you, continue to hone your skills and dig deeper into the topics covered in this module and course - we can never learn too much. Best of luck!

You've completed:
Unsupervised learning